PAHs near Cosmic Noon: Slit-Stepping with JWST/MIRI LRS

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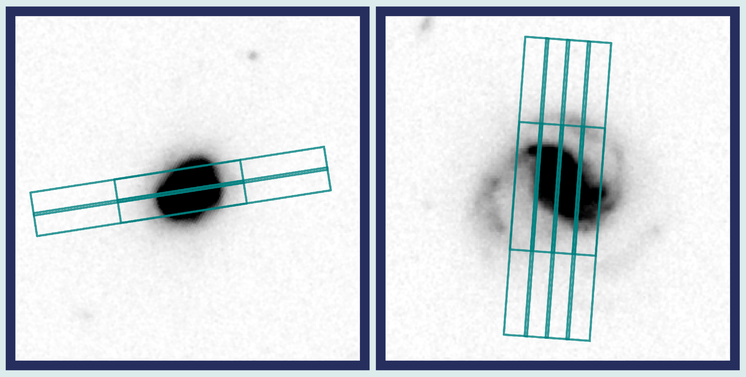
Introduction

JWST provides an unprecedented opportunity to explore ISM conditions beyond the local Universe, as the MIRI instrument now permits the detection of the 3.3 μm PAH feature for large samples of galaxies beyond $z = 0$. The spatial resolution of the MIRI Low-Resolution Spectrograph also now allows new, spatially resolved studies of the 3.3 μm PAH and other, more well-calibrated PAH features at high z. To take advantage of these capabilities and explore the correlation of PAHs and molecular gas near cosmic noon, a non-standard approach to MIRI/LRS observational design and data reduction is required.



Left: Slit layouts for two galaxies: one more compact (one slit, left) and one very extended (four slits, right).

Right: Three snapshots of the data cube for a $z = 0.70$ galaxy, at (left to right) rest-frame 3.3, 4.6, and 6.2 μm.

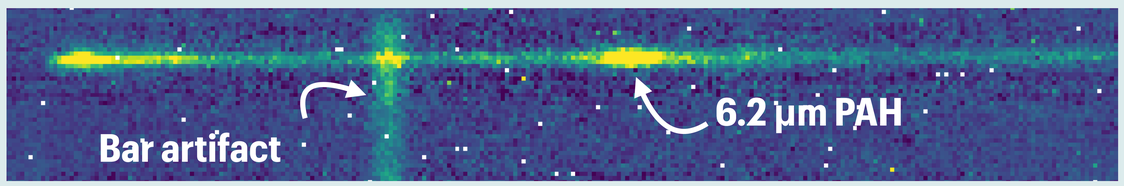
Observational Design

Why MIRI/LRS?

- Higher sensitivity than MIRI/MRS for efficient observations: a statistical sample (22 galaxies) in <12 hours science time
- Sufficient resolution for detailed study of broad PAH features

Design Considerations

- Mosaicing of LRS slit: PA restrictions for elongated targets
- Design observations to create pseudo-IFU: morphological information for distribution of PAHs vs. continuum, lines



Above: Background-subtracted 2D spectrum from a single nod with the bright bar artifact visible.

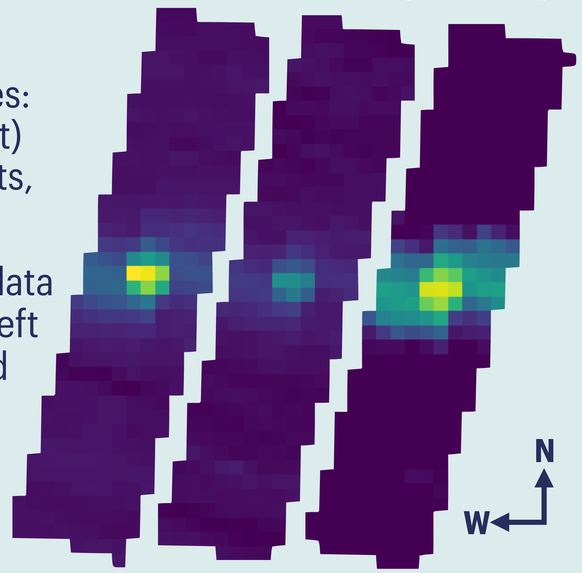
Right: Spectrum of the $z = 0.70$ galaxy CS889164, extracted from the data cube. The 3.3, 5.3, 5.7, and bright 6.2 μm complexes are visible.

Data Reduction

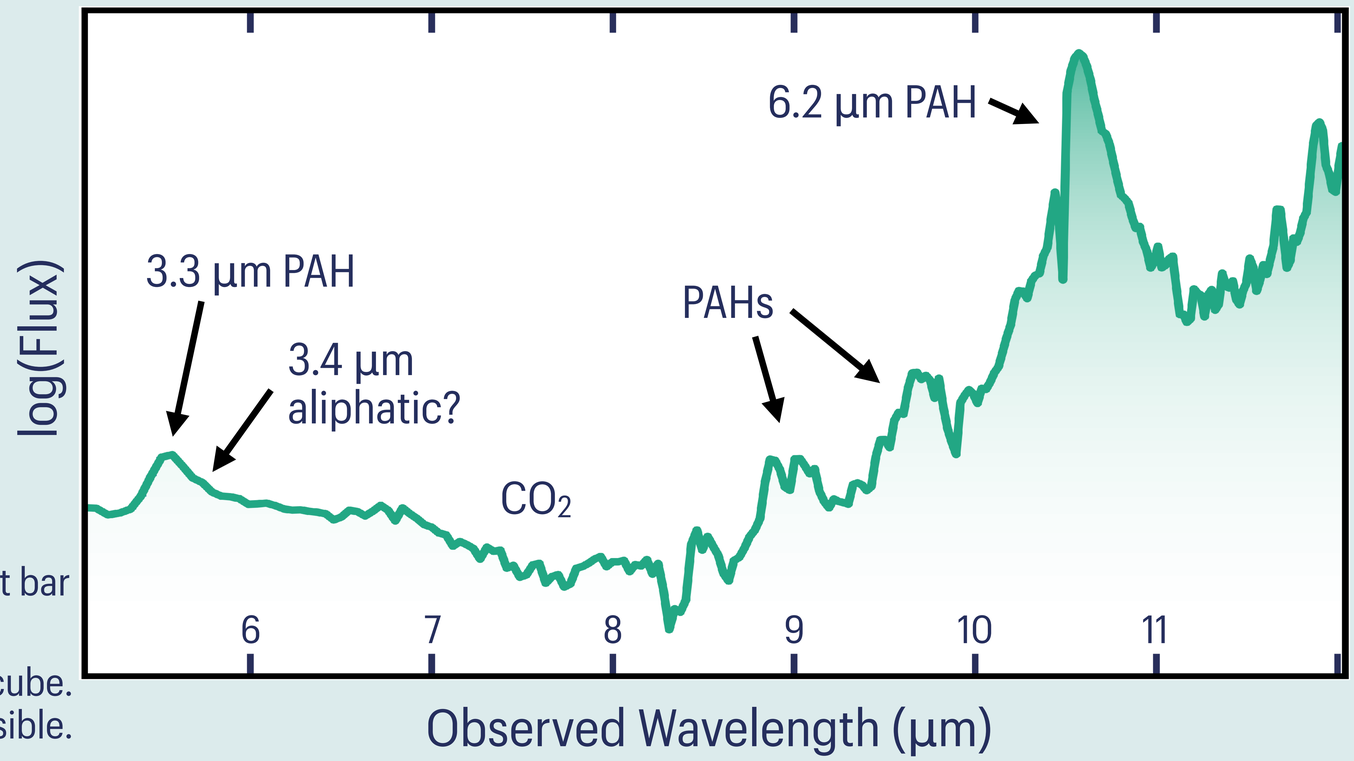
Background Subtraction

- Combine observations close on sky & in time (within ~1 month) for most robust background
- Bar-shaped persistence artifact near 8.3 μm observed-frame decays across successive LRS observations
- Bar always present if TA is performed; must correct separately from normal sky background

Astrometry correction using TA images necessary for pre-August 2025 observations to correct pointing uncertainty and center source in cube



LRS Cube Building

We developed custom routines based on the JWST MIRI IFU pipeline to build data cubes. In short, we:

- map each slit to the sky using the WCS
- set up the cube based on the total footprint of the observations
- drizzle in the slits to fill in the flux and variance of each spaxel
- Update the WCS and return the final cube